

SIM2.2 — 035 NMOS Channel-Length Modulation

Setup and selected bias

This simulation uses the NM Level 8 BSIM3 model and nmos_035 symbol from the supplied 035 LTspice library. The source and bulk are grounded. The aspect ratio is held at $W/L = 10$ for all five devices.

The assignment requires a constant overdrive but does not prescribe its numerical value or the drain-voltage sweep endpoints. We choose a nominal $V_{OV} = 0.500$ V as a clear, practical bias point. For the supplied model, the threshold at the 1 V reference point is about 0.500–0.536 V across the five lengths, so this target gives V_{GS} values near 1.0 V. V_{GS} is calibrated separately for each length from the LTspice operating-point V_{th} at $V_{DS} = 1.0$ V, using $V_{GS} = V_{th} + V_{OV}$; this keeps the nominal overdrive comparable as L changes.

L (μm)	W (μm)	Vth at VDS = 1 V (V)	VGS (V)	Nominal VOV (V)
1	10	0.536	1.036	0.500
2	20	0.518	1.018	0.500
5	50	0.506	1.006	0.500
10	100	0.502	1.002	0.500
20	200	0.500	1.000	0.500

The full characteristic is swept from $V_{DS} = 0$ V to 2.0 V in 10 mV steps. This displays the low- V_{DS} triode region as well as the saturation region. The upper endpoint of 2.0 V is a selected sweep limit that gives a broad drain-voltage range for observing the output slope.

With the simple long-channel criterion, the saturation boundary is $V_{DS} \approx V_{GS} - V_{th} = V_{OV}$, nominally about 0.5 V here. We use the interval from 0.5 V to 2.0 V for the λ calculation, starting at the selected nominal boundary. Points below 0.5 V remain in the full-characteristic plot but are excluded from the saturation calculation.

The gate voltages are fixed during each drain sweep. Because the BSIM3 model can shift its effective V_{th} slightly with V_{DS} , the 0.500 V overdrive is matched at the 1.0 V reference point and is nominal over the sweep rather than mathematically exact at every point.

Method

In saturation, use the assignment's approximation:

$$I_D \approx I_{D0}(1 + \lambda V_{DS})$$

Estimate the average slope over the selected 0.5–2.0 V interval, using the simulated values at its two endpoints:

$$\begin{aligned} g_{ds} &\approx [I_D(2.0 \text{ V}) - I_D(0.5 \text{ V})]/(2.0 - 0.5 \text{ V}) \\ I_{D0} &= I_D(0.5 \text{ V}) \\ \lambda &\approx g_{ds}/I_{D0} \end{aligned}$$

The assignment does not specify which saturation-region bias point should be used for I_D in the r_o formula. Here we choose $V_{DS} = 1.5 \text{ V}$ because it is a round value inside the 0.5–2.0 V saturation interval. A different saturation-region V_{DS} would give a slightly different r_o , because $r_o = 1/(\lambda I_D)$ depends on the bias current.

Then calculate output resistance at the selected $V_{DS} = 1.5 \text{ V}$ using the requested formula:

$$r_o \approx 1/[\lambda I_D(1.5 \text{ V})]$$

The sweep starts at 0 V for the plot, but samples below 0.5 V are not used in these saturation-region calculations.

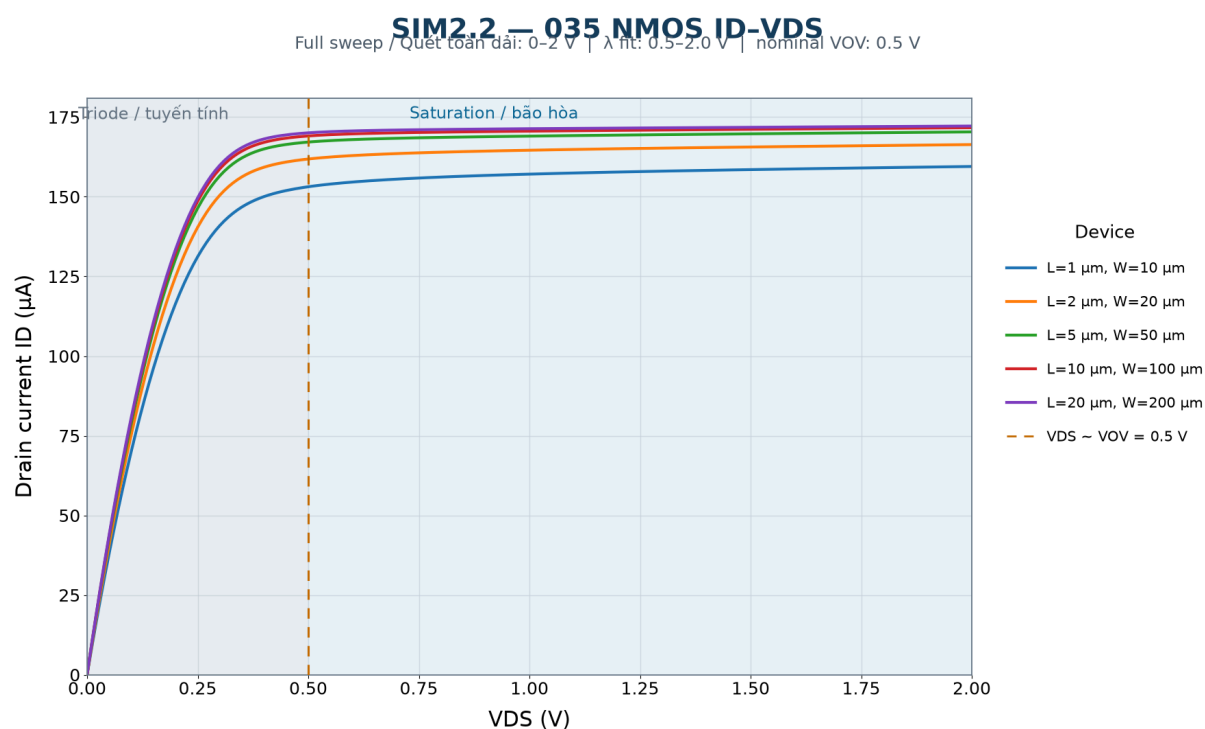
Results

L (μm)	W (μm)	I_D at 1.5 V (μA)	Slope g _{ds} (μS)	λ (V ⁻¹)	r_o (kΩ)
1	10	158.519	4.238	0.027670	227.985
2	20	165.615	3.017	0.018642	323.898
5	50	169.750	2.130	0.012743	462.306
10	100	171.163	1.727	0.010215	571.923
20	200	171.800	1.422	0.008362	696.075

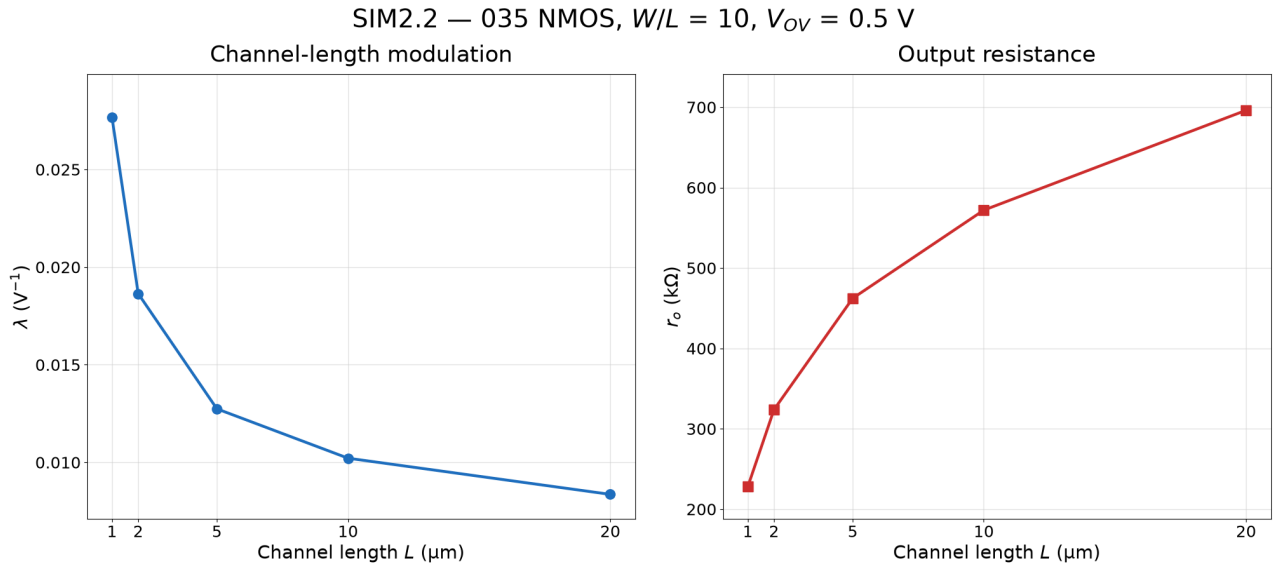
Plots

Full drain-current characteristic, from 0 V through saturation

The shaded low-voltage region shows the triode portion. The blue shading marks the 0.5–2.0 V interval used to estimate λ .



Channel-length modulation and output resistance versus channel length



Discussion

The full I_D – V_{DS} plot shows the nearly linear rise at low V_{DS} , followed by the flatter saturation-region curves. Only the latter interval is used to estimate channel-length modulation, as required by the saturation approximation.

As L increases from $1\ \mu\text{m}$ to $20\ \mu\text{m}$, λ decreases from $0.027670\ \text{V}^{-1}$ to $0.008362\ \text{V}^{-1}$, while r_o increases from $228.0\ \text{k}\Omega$ to $696.1\ \text{k}\Omega$. The output curves become flatter for longer devices, showing weaker channel-length modulation. The current at $1.5\ \text{V}$ rises modestly from $158.5\ \mu\text{A}$ to $171.8\ \mu\text{A}$ even though W/L and nominal overdrive are held constant. This is the behavior of the supplied BSIM3 model across these geometries; the ideal long-channel square-law model would predict less variation.

Files

- ../Simulation/SIM2_2_NMOS.asc — LTspice schematic; sweeps the full V_{DS} range from 0 to 2.0 V and measures λ and r_o over the saturation interval.
- ../Simulation/SIM2_2_NMOS_VTH.cir — operating-point threshold calibration at $V_{DS} = 1\ \text{V}$.
- ../Simulation/SIM2_2_NMOS_ID_VDS.csv — full 0–2 V output-characteristic data.
- ../Simulation/SIM2_2_NMOS_results.csv — metrics calculated from the 0.5–2.0 V saturation interval.